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ABICHU CAMPUS

DEPARTMENT OF COMPUTER SCIENCE

COMPUTER SECURITY
NETWORK SECURITY

A Beginner-to-Presentation Study Book

Group One

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Prepared for the 30-minute group presentation + 10-minute Q&A

How to Use This Book

This short textbook is designed to teach Group One Network Security from the beginning. It follows the exact presentation areas: network security basics, network threats, trust and risk, TCP/IP weaknesses and buffer overflows, security protocols by layer, physical security, and wireless security.

Do not try to memorize the slides. First understand the ideas, then practice explaining them in your own words. Each chapter contains learning objectives, simple explanations, examples, key terms, and a short self-test.

Recommended method: learn one chapter, close the book, explain the topic aloud, then answer the self-test.

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1. What Is a Computer Network?

A computer network is a collection of computers and other devices connected so they can communicate and share resources. Examples include a university campus network, a home Wi-Fi network, and the Internet.

1.1 Main Network Components

Host: A device that communicates on a network, such as a laptop, phone, or server.

Switch: Connects devices within a local network and forwards Ethernet frames.

Router: Connects different networks and forwards IP packets between them.

Access Point: Provides wireless devices with network connectivity.

Server: Provides a service, such as web, DNS, e-mail, or file storage.

Client: A device or application that requests a service.

1.2 Why Networks Need Security

Networks carry valuable information: passwords, student records, financial information, business documents, and application traffic. An attacker who gains unauthorized access may steal information, change it, or make services unavailable.

Quick Check

1. What is the difference between a switch and a router?
2. Give three examples of network hosts.
3. Why is a university network a security target?

2. What Is Network Security?

Network security is the collection of technologies, configurations, policies, and practices used to protect networked systems and communications from unauthorized access, misuse, modification, disruption, and destruction.

2.1 Security Controls

Firewall: Controls network traffic according to security rules.

IDS: Intrusion Detection System; detects suspicious activity and generates alerts.

IPS: Intrusion Prevention System; detects and can block malicious activity.

Authentication: Verifies who a user or device is.

Authorization: Determines what an authenticated subject is allowed to do.

Encryption: Transforms readable data into protected ciphertext using cryptographic methods.

Network segmentation: Separates a network into zones so compromise of one area does not automatically expose everything.

2.2 Defense in Depth

A secure network should use multiple layers. For example, a firewall may block unwanted traffic, authentication limits access, encryption protects data in transit, endpoint security protects devices, monitoring detects suspicious behavior, and backups support recovery.

Quick Check

4. What is the purpose of a firewall?
5. What is the difference between IDS and IPS?
6. Why is using several security controls better than relying on one?

3. The CIA Triad and Security Goals

The CIA Triad is the foundation of information security: Confidentiality, Integrity, and Availability.

Principle	Meaning	Example
Confidentiality	Only authorized parties can access information.	Encryption protects a student's records.

Integrity	Information remains accurate and is not changed without authorization.	A digital signature can reveal modification.
Availability	Authorized users can access systems when needed.	Redundant servers help keep a website online.

Memory trick: Confidentiality = who can SEE it; Integrity = who can CHANGE it; Availability = who can USE it.

A DDoS attack primarily affects Availability. A stolen database primarily threatens Confidentiality. Unauthorized modification threatens Integrity.

4. Network Architecture and the TCP/IP Model

The TCP/IP model organizes network communication into layers. Different protocols perform different jobs at different layers.

Layer	Examples	Main job
Application	HTTP/HTTPS, DNS, SMTP	Provides services used by applications.
Transport	TCP, UDP, TLS operates above transport in the common Internet stack	Provides end-to-end transport functions.
Internet	IP, ICMP	Provides addressing and routing between networks.
Link	Ethernet, ARP, Wi-Fi	Moves frames across a local network/link.

Remember that security can also be layered. HTTPS protects application communication, TLS protects transport-level communication, IPsec protects IP communication, and WPA3 protects wireless links.

5. Threats on Networks

A network threat is a potential event, actor, or technique that can cause harm to networked resources.

Malware: Malicious software. Worms can spread automatically; ransomware can encrypt data; spyware can collect information.

DoS/DDoS: Attempts to make a service unavailable. DDoS uses multiple sources.

MitM: An attacker gets between communicating parties and attempts to observe or alter communication.

Packet sniffing: Capturing packets from network traffic. Encryption reduces the value of captured traffic.

Spoofing: Pretending to be another system or identity, such as ARP or IP spoofing.

Phishing: Social engineering used to trick people into opening content or giving credentials.

Credential attacks: Attempts to obtain passwords, tokens, or other authentication secrets.

Session hijacking: Taking over an already authenticated session, often by stealing a session token.

Attack vs. Defense

Security professionals should think in stages: How does the attacker enter? What can they execute? What can they steal? Can they move laterally? Can monitoring detect them? Which control can stop each stage?

6. Trust, Weaknesses, Threats, Vulnerabilities, and Risk

These terms are related but not interchangeable. Understanding their differences is essential for exams and for explaining security incidents.

Trust: A decision to rely on a user, device, service, or network based on evidence or policy.

Weakness: A flaw or condition that may reduce security.

Vulnerability: A weakness that can be exploited to cause harm.

Threat: A potential source or cause of harm.

Exploit: A technique or code that takes advantage of a vulnerability.

Impact: The damage or loss caused by a successful incident.

Risk: The potential for loss when threats can exploit vulnerabilities; likelihood and impact are important factors.

Example: An unpatched web server is a vulnerable asset. A criminal group is a threat. Exploiting the vulnerability is an attack. If the server contains sensitive records, the impact may be high, making the risk significant.

Risk Reduction

- Reduce likelihood: patch systems, restrict access, harden configurations.
- Reduce impact: backups, segmentation, least privilege, incident response.
- Transfer/share risk: insurance or contracts where appropriate.
- Accept risk: only when the organization consciously decides the remaining risk is acceptable.

7. TCP/IP Suite Weaknesses

Many classic Internet protocols were designed when the Internet was smaller and more trusted. Security mechanisms were often added later rather than being built into every protocol.

7.1 IP Weaknesses

Basic IP provides addressing and routing but does not by itself provide confidentiality. Source addresses can also be spoofed.

7.2 ARP Weaknesses

ARP maps IP addresses to MAC addresses on local networks. Traditional ARP does not strongly authenticate ARP messages, enabling ARP spoofing/poisoning.

7.3 DNS Weaknesses

Traditional DNS was not designed to encrypt all queries or strongly authenticate every response. DNS security extensions and encrypted DNS technologies address different parts of the problem.

7.4 TCP Weaknesses

TCP provides reliable transport but does not itself encrypt application data. Secure protocols such as TLS can be used above TCP.

Important idea: A protocol can work correctly and still be insecure if an attacker can abuse assumptions made by the protocol or its environment.

8. Buffer Overflow and Memory Protection

A buffer is a memory area used to hold data. A buffer overflow occurs when software writes beyond the buffer's intended boundary. The excess data may overwrite neighboring memory, potentially causing a crash or altering control data.

8.1 Why It Matters

Depending on the program and protections, memory corruption can lead to denial of service or, in serious cases, arbitrary code execution.

8.2 Main Mitigations

Bounds checking: Make sure programs do not write more data than a buffer can safely hold.

Stack Canaries: Place a special value near sensitive stack control data. Unexpected modification can reveal an overflow before returning from a function.

ASLR: Randomizes important memory locations, making it harder to predict where useful code or data will be located.

DEP/NX: Marks selected memory areas as non-executable, making simple code-injection attacks harder.

Memory-safe languages: Languages with strong memory-safety guarantees can prevent many classes of memory corruption.

Exam distinction: ASLR randomizes locations; DEP/NX blocks execution from protected memory; Stack Canaries detect corruption of the canary.

9. Network Security Protocols

Security protocols are standardized mechanisms that provide properties such as confidentiality, integrity, authentication, and secure access.

Security by Layer

Application: HTTPS/TLS for web communication; secure e-mail mechanisms and domain authentication.

Transport: TLS protects application communication over networks.

Network: IPsec can protect IP traffic.

Link: 802.1X, WPA2/WPA3 and other local-link controls.

Physical: Locks, access controls, CCTV, power and environmental protections.

10. Application-Layer Security: Web Security

Web applications receive untrusted input from users and communicate across networks, so they need both application security and transport security.

HTTPS

HTTPS is HTTP carried over TLS. It protects web traffic from network eavesdropping and helps authenticate the server.

Common Web Security Controls

- Parameterized SQL queries/prepared statements to reduce SQL Injection risk.
- Output encoding and safe DOM practices to reduce XSS.
- Anti-CSRF protections for state-changing requests.
- Secure, HttpOnly, and appropriate SameSite cookie settings.
- Strong authentication and authorization checks.
- Server-side validation; never trust client-side validation alone.
- Least privilege for application and database accounts.

Important distinction: HTTPS protects data in transit. It does not automatically make a vulnerable application secure.

11. Application-Layer Security: E-mail Security

E-mail is attacked through phishing, malware attachments, spoofing, credential theft, and interception.

Important Technologies

TLS: Encrypts communication between mail systems when both sides support it.

SPF: Publishes authorized sending sources for a domain.

DKIM: Uses a cryptographic signature to help verify that an email was authorized by the sending domain and not altered in transit.

DMARC: Provides a policy and reporting framework using domain alignment with SPF and/or DKIM.

These mechanisms complement each other. They do not replace user awareness, malware filtering, or strong authentication.

12. Transport-Layer Security: TLS

TLS (Transport Layer Security) creates a protected communication channel. It is widely used by HTTPS and other Internet applications.

TLS 1.3 Simplified Handshake

7. ClientHello: browser sends supported parameters and a key share.
8. ServerHello: server selects parameters and sends its key share.
9. Key agreement: both sides derive shared secret material using ephemeral key exchange.
10. Certificate: server sends a certificate that binds its identity to a public key.
11. Authentication: server proves possession of the private key by signing handshake data.
12. Verification: browser validates the certificate chain and signature.
13. Key derivation: both sides derive symmetric session keys.

14. Encrypted communication: application data is protected using the established keys.

Why symmetric keys after the handshake? Symmetric encryption is efficient for protecting large amounts of data. Public-key mechanisms are mainly used for authentication and key establishment.

13. Network-Layer Security: IPsec

IPsec is a suite of protocols and mechanisms for protecting IP communications. It can provide authentication, integrity, confidentiality, and anti-replay protection depending on configuration.

Common Uses

- Site-to-site VPNs connecting organizational networks.
- Remote-access VPNs in some architectures.
- Protection of IP traffic between selected hosts or gateways.

Key Idea

IPsec operates at the IP layer, so applications do not necessarily need to be individually modified to benefit from protected IP traffic.

14. Link-Layer Security: Ethernet, 802.1X, and Wi-Fi

The link layer handles communication on a local network segment. Security at this layer can prevent unauthorized devices from gaining network access and can protect wireless traffic.

802.1X

IEEE 802.1X is a port-based network access control framework commonly used with enterprise authentication. It can require a device/user to authenticate before obtaining normal network access.

WPA2 and WPA3

WPA2 and WPA3 are Wi-Fi security standards. WPA3 provides modern security improvements and should be preferred where supported and correctly configured.

Do not confuse Wi-Fi security with Internet security: securing the wireless link does not automatically secure every application or service accessed through it.

15. Physical Security

Physical security protects the actual hardware and facilities that support a network. Cybersecurity can fail if an attacker can simply walk into a server room and access equipment.

- Locked server and network rooms.
- Badge, biometric, or other controlled entry.
- CCTV and security monitoring.
- Secure racks and cable management.
- UPS and backup power.
- Fire detection/suppression and environmental monitoring.
- Asset inventory and secure disposal of equipment and storage media.

Physical security supports all three CIA goals. Theft can expose data, tampering can change systems, and power/environmental failures can reduce availability.

16. Wireless Security

Wireless networks communicate using radio signals, so an attacker may be able to interact with the network without having physical access to a cable.

Common Wireless Risks

Weak passwords: Attackers may guess or obtain credentials.

Rogue access points: Unauthorized APs can be used to deceive users or bypass security.

Evil-twin attacks: A fake AP imitates a legitimate network name.

Eavesdropping attempts: Attackers may try to capture wireless traffic.

Misconfiguration: Insecure protocols, default passwords, or poor segmentation increase exposure.

Wireless Protection

- Prefer WPA3 where practical; use enterprise authentication for organizational networks when appropriate.
- Use strong, unique credentials and change default administrative passwords.
- Keep access-point firmware updated.
- Separate guest Wi-Fi from internal systems.
- Disable obsolete/insecure protocols and unnecessary management services.
- Monitor for unauthorized access points.
- Use HTTPS for applications and a trusted VPN when appropriate on untrusted networks.

17. Defense in Depth and Secure Network Design

Defense in Depth means applying several independent or complementary controls so that failure of one control does not expose the entire organization.

Example

A university network might use:

- Internet firewall
- Network segmentation between students, staff, servers, and guests
- MFA for administrators
- WPA3/enterprise Wi-Fi authentication
- HTTPS/TLS
- Endpoint protection
- IDS/IPS and centralized logging
- Regular patching and vulnerability management
- Backups and incident response

This is stronger than saying 'we have a firewall, so we are secure.' No single control is perfect.

18. Putting Everything Together: A University Network

Imagine a university with student Wi-Fi, staff computers, web servers, an e-mail service, and an administrative database.

Possible Security Design

Perimeter: Firewall and DDoS protection help control Internet-facing traffic.

Network zones: Separate student, staff, server, and guest networks.

Identity: Use strong authentication and MFA for sensitive accounts.

Web: HTTPS, secure sessions, authorization checks, input validation, and secure database access.

E-mail: TLS plus SPF, DKIM, DMARC, filtering, and phishing awareness.

Wi-Fi: WPA3/enterprise authentication and guest isolation.

Monitoring: IDS/IPS, logs, and alerting help identify attacks.

Physical: Secure server rooms, network equipment, power, and environmental controls.

Recovery: Tested backups and incident response procedures.

Attack Scenario

Suppose an attacker connects to a fake Wi-Fi access point and tries to intercept a student's traffic. HTTPS protects properly configured web traffic in transit. Network segmentation limits access to internal systems. Strong authentication reduces account compromise. Monitoring may detect unusual behavior. This demonstrates how multiple controls work together.

19. Presentation Preparation and Q&A

How to Present

Do not read every word on the slide. Use the slide as a visual guide and explain the concept naturally. Each member should be able to define their topic, give one example, name one attack or weakness, and explain at least one defense.

Suggested 5-Minute Structure for Each Member

15. 30 seconds: introduce the topic and explain why it matters.
16. 2 minutes: explain the core concepts.
17. 1 minute: give a concrete real-world example.
18. 1 minute: explain defenses or security controls.
19. 30 seconds: summarize and transition to the next member.

Questions You Should Be Ready For

- What is the CIA Triad and why is it important?
- What is the difference between a threat and a vulnerability?
- Why does HTTPS not make a vulnerable web application secure?
- What is the difference between IDS and IPS?
- How is a DDoS attack related to Availability?
- Why is ARP vulnerable to spoofing?
- What does TLS protect?
- What is the difference between TLS and IPsec?
- Why is WPA3 better than using an open Wi-Fi network?
- What is a buffer overflow and how do Stack Canaries, ASLR, and DEP differ?
- Why is defense in depth important?

20. Final Review Quiz

1. What are the three components of the CIA Triad?

Answer: Confidentiality, Integrity, Availability.

2. Which CIA property is primarily affected by DDoS?

Answer: Availability.

3. What does a firewall do?

Answer: Controls network traffic according to security rules.

4. What is a vulnerability?

Answer: A weakness that can be exploited.

5. What is ARP spoofing?

Answer: Sending forged ARP information to manipulate IP-to-MAC mappings on a local network.

6. What is a buffer overflow?

Answer: Writing beyond a buffer's intended memory boundary.

7. What does ASLR do?

Answer: Randomizes important memory locations.

8. What does DEP/NX do?

Answer: Prevents execution from protected/non-executable memory regions.

9. What does HTTPS use to secure web traffic?

Answer: TLS.

10. What does IPsec protect?

Answer: IP-layer communication, depending on configuration.

11. What is 802.1X used for?

Answer: Network access control/authentication at the link layer.

12. Name two Wi-Fi security standards.

Answer: WPA2 and WPA3.

13. What is a MitM attack?

Answer: An attacker positions themselves between communicating parties to intercept or manipulate traffic.

14. Why is physical security part of network security?

Answer: Because attackers can steal, tamper with, or disable network equipment and data.

15. What is Defense in Depth?

Answer: Using multiple layers of security controls so one failure does not compromise everything.

Quick Glossary

Term	Meaning
AP	Access Point
ARP	Address Resolution Protocol
CIA	Confidentiality, Integrity, Availability
DDoS	Distributed Denial of Service
DNS	Domain Name System
IDS	Intrusion Detection System

IPS	Intrusion Prevention System
IPsec	Internet Protocol Security
MitM	Man-in-the-Middle
NAT	Network Address Translation
TCP	Transmission Control Protocol
TLS	Transport Layer Security
UDP	User Datagram Protocol
VPN	Virtual Private Network
WPA	Wi-Fi Protected Access